TADA! Adventure Map

The TADA! Adventure Map

(The Tangible Art and Design Adventure Map)

Introductory Points of Interest

Activity 1: Introduction to Tactile Graphics

Introduction to tactile graphics and art, with an emphasis on hands-on learning and creativity while introducing the idea of using technology to create tactile graphics.

Goals: Develop motor skills, spatial awareness, and appreciation for accessible art.

Activity 2: Descriptive Language in Drawing

Explore descriptive language and tactile graphics. Practice developing directions for replication.

Goal: Learn importance of using precise directions when creating tactile graphics.

Activity 3: Tactile Colors

Develop strategies for creating pixel art using a grid, braille, and textured paper. Practice how to use a key to "color-code" a graphic.

Goals: Learn about color and develop tactile discrimination skills.

Activity 4: Moving from 3D to 2D: Basic Shapes

The translation from 3D to 2D representations benefits from direct, explicit instruction to help students develop a strong foundation for interpreting tactile graphics. **Goals:** Learn how 3D objects are represented as 2D tactile graphics.

Activity 5: Coming Soon

Activity 6: Coming Soon

Intermediate Points of Interest

Activity 1: Code & Go Mouse

Play with Colby the programmable robot mouse from APH while incorporating spatial concepts in real life. Practice sequencing and mapping while having fun.

Goals: Learn basic coding skills and exercise computational thinking.

Activity 2: Colby's Mouse Town

Plan and navigate routes through Colby's town on graph paper. Practice sequencing by breaking bigger tasks into smaller steps.

Goal: Learn foundational programming concepts.

Activity 3: Drawing with Colby

Plan and use Colby's path to create shapes. Use coordinates on a coordinate grid as a canvas.

Goals: Introduce perspective and use of a coordinate grid; practice introductory programming concepts.

Activity 4: CodeQuest

Learn coding skills with CodeQuest, a free iPad app from APH. Program solutions to help the astronaut return to their spaceship.

Goals: Apply new coding skills to complete an adventure. Use a computer with a text editor program; open and navigate .svg files, alter .svg code to achieve desired outcomes; build confidence with troubleshooting and iterative design processes.

Advanced Points of Interest

Intro to SVG

Activity 1: Intro to Spatial Mapping

Practice using a coordinate plane to map SVG canvas units to paper. Draw basic shapes on graph paper and explore how to mark and measure shapes.

Goals: Introduce a coordinate system for graphing points. Apply spatial reasoning to shapes and dimensions on a coordinate plane.

Activity 2: Digitizing Drawings with SVG Code

Learn SVG code basics. Translate physical drawings to digital illustrations. Experiment with different ways of rendering digital illustrations as tactile graphics.

Goals: Use a computer with a text editor program; learn how to render and open .svg files; use an embosser or other accessible rendering tool to create tactile graphics from digital media.

Activity 3: Iterating on SVG Drawings and Beyond

Iterate (repeatedly produce) digital designs, change shapes as desired, and practice using SVG syntax to prepare for more advanced coding and drawing skills. Use rendered tactile graphics as a means to draft and revise designs.

Moving from 2D to 3D design

Section A: Making 2D Shapes in OpenSCAD

- Activity A.1. Create a Square in OpenSCAD
- Activity A.2. Basic Shapes in OpenSCAD
- Activity A.3. Rotating, Translating, Scaling and Mirroring Shapes

Programming in OpenSCAD — Section B

- Activity B.1. Adding and Subtracting Objects
- Activity B.2. Moving from 2D to 3D
- Activity B.3. True 3D Models

Section C: Guide to Published 3D Printable STEM Models

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